

Steam Systems

Three of your repeated misses are steam questions, and they all turn on one piece of equipment: the water level gauge.

A steam boiler only fills partway. It needs water at the bottom to boil and space above it for the steam to collect, so the water level is something that has to be watched — and that is the giveaway for the whole system type.

YOU HAVE MISSED THIS 3 TIMES

Which of the following would indicate that the boiler is being used for a steam heating system rather than a hot water heating system?

There is a water level gauge on the boiler.

A hot water boiler is completely full of water under pressure, so there is no level to see and no gauge to show it. A sight glass on the side of the boiler means steam.

YOU HAVE MISSED THIS 4 TIMES

Problems are evident when the water level gauge in a steam heating system:

All of the answers.

Missed four times, and the reason is the shape of the question rather than the steam. The options are separate faults — the glass is dirty or cloudy, the level is too high, the level is too low, the level does not move — and none of them excludes the others. Separate faults accumulate, so "all" is live. Ask whether the options compete or add up.

YOU HAVE MISSED THIS 4 TIMES

In what location would an equalizer pipe be found?

Near a steam boiler.

Equalizer pipe

Connects the steam side to the return near a steam boiler, keeping the pressure on both sides balanced so steam pressure cannot push the boiler water out into the return.

How steam is conveyed

By the boiler and pipes. Steam travels under its own pressure with no pump at all — which is why the condensate has to return by gravity, and why steam piping must be pitched.

YOU HAVE MISSED THIS

A steam heating system is a type of:

Hydronic system.

Say it back

Cover the page. Answer each one out loud.

1. What tells you a boiler is serving a steam system rather than hot water?
2. Why does a hot water boiler have no water level gauge?
3. Name the faults a steam water level gauge can show, and say why the answer is cumulative.
4. Where is an equalizer pipe found, and what does it do?
5. What conveys heat to the radiators in a steam system?
6. Why must steam piping be pitched?